Roll Number: _____

Thapar Institute of Engineering and Technology,
Patiala Department of Computer Science and Engineering

B E- COE, CSE (Summer Sem)
MST Course Code: UCS701 — Course Name: Theory of Computation
July 16th, 2026, 9.00 AM — Time: 3 Hours, M. Marks: 40
Name Of Faculty: Dr. Nidhi Chakravarty

Note: Attempt any 5 questions with proper justification. Assume missing data, if any, suitably.

S.no	Questions	Marks
Q1	In each case below, find a CFG generating the given language. **a.** The set of odd-length strings in {a,b} with middle symbol a. **b.** The set of even-length strings in {a,b} with the two middle symbols equal. **c.** The set of odd-length strings in {a,b} whose first, middle, and last symbols are all the same. **d.** Find a context-free grammar generating the language $\{a^i b^j c^k \mid i \neq j + k\}$	8
Q2	Simplify the following CFG. $S \rightarrow aA \mid aBB$ $A \rightarrow aAA \mid \varepsilon$ $B \rightarrow bB \mid bbC$ $C \rightarrow B$	8
Q3	Convert to Chomsky Normal Form (CNF) $S \rightarrow AACD$ $A \rightarrow aAb \mid \varepsilon$ $C \rightarrow aC \mid a$ $D \rightarrow aDa \mid bDb \mid \varepsilon$	8
Q4	**a)** Construct a context-free grammar over {0,1} which generates all strings with equal number of 0's and 1's (Few accepted strings for example 0110,1010, 001110,...}. Justify your answer. b) Consider the following grammar and convert it into equivalent GNF grammar. $S \rightarrow AB$ $A \rightarrow BS \mid b$ $B \rightarrow SA \mid a$	2+ 6
Q5	Given the context-free grammar $S \rightarrow XY$ $X \rightarrow 0 \mid YY$ $Y \rightarrow 1 \mid XY$ Apply CYK algorithm to determine whether the string w=00111 belongs to the language generated by above given grammar.	8
Q6	a) Write down the logic for the design of a Turing machine and design the Turing Machine for the above designed logic for the language $L = \{a^n b^{2n} \mid n > 0\}$	3+5
